1. Project Idea

FixFlow AI is a web-based intelligent maintenance management system designed for organizations that receive frequent maintenance and service requests.

Instead of relying on phone calls, WhatsApp messages, paper forms, or spreadsheets, FixFlow AI provides a centralized platform where users can:

- Report problems
- Upload photos
- Specify locations
- Track request status
- Receive notifications

Meanwhile, managers can:

- Review requests
- Automatically classify problems using AI
- Identify severity
- Assign technicians
- Monitor repair progress
- Analyze maintenance performance

(The system is designed as a commercial B2B SaaS product, meaning it can potentially be sold to different organizations)

2. Problem Statement

Organizations such as offices, apartment buildings, hotels, schools, hospitals, factories and shopping centers face maintenance problems every day.

Examples include:

- Broken air conditioners
- Electrical problems
- Water leakage
- Plumbing problems
- Damaged furniture
- Broken doors/windows
- Network problems
- Equipment failures
- Cleaning issues
- Safety hazards

In many organizations, these problems are reported informally through phone calls, messaging applications or verbal communication.

This creates several difficulties.

Current problems

1. Maintenance requests can be lost
A request made verbally or through a message may not be properly recorded.

2. Difficult to prioritize
A minor broken chair and a dangerous electrical problem may not be distinguished properly.

3. Wrong technician assignment
A manager may need to manually determine which technician has the appropriate skills.

4. Lack of transparency
The person who reported a problem often doesn't know whether anyone is working on it.

5. Repeated problems
The same equipment may repeatedly fail without management noticing the pattern.

6. Lack of performance information

Management may not know:

- How many problems occur?
- Which location has the most problems?
- Which type of maintenance is most common?
- How long does a repair usually take?
- Which equipment fails repeatedly?

3. Proposed Solution

FixFlow AI will provide a centralized maintenance management platform.

Basic workflow

User notices a problem
↓
Reports the problem
↓
AI analyzes the report
↓
Category + severity suggested
↓
Duplicate request checked
↓
Manager reviews request
↓
Suitable technician assigned
↓
Technician performs repair
↓
Technician updates status
↓
User verifies the repair
↓
Ticket closed

The system will maintain the entire history of the maintenance request.

4. Example Real-Life Scenario

Suppose an employee notices:

"The AC in Meeting Room B is running but the room is not cooling."

The employee opens FixFlow AI and submits:

Category: Automatically detected
Location: Building A → Floor 3 → Meeting Room B
Description: AC running but not cooling
Photo: Attached

The AI analyzes the description and recommends:

Problem Category: HVAC
Severity: Medium
Confidence: 91%

The system then checks whether another unresolved request already exists for the same location and problem.

If not, it creates:

Ticket #FF-1024

The manager sees the request on the dashboard.

The system recommends an available HVAC technician.

The manager approves the assignment.

The technician receives:

New maintenance request — Meeting Room B

After repairing the AC, the technician changes:

Assigned
↓
In Progress
↓
Resolved

The employee receives a notification:

"Your maintenance request has been resolved. Please verify the repair."

If the AC is still not working, the employee selects:

Problem Not Fixed

The ticket becomes:

Resolved → Reopened

The manager is notified.

This creates a complete, traceable workflow.

5. Vision Statement

To provide organizations with an intelligent, transparent and efficient maintenance management platform that reduces response time, minimizes unresolved service requests and enables data-driven maintenance decisions.

6. Target Users

The product is intentionally designed for multiple types of customers.

Potential business customers

- Office buildings
- Apartment management companies
- Hotels
- Restaurants
- Schools and universities
- Hospitals and clinics
- Shopping malls
- Factories
- Warehouses
- Property management companies

System users

There are four primary roles.

1. Requester
Employees, tenants, students, customers or other authorized users who report problems.

2. Maintenance Technician
Employees responsible for solving maintenance problems.

3. Manager
Responsible for reviewing, prioritizing and assigning maintenance requests.

4. Organization Administrator
Responsible for users, locations, categories, system configuration and organizational settings.

7. Project Objectives

The main objectives are:

- Create a centralized platform for maintenance requests.
- Reduce the time required to report and assign maintenance problems.
- Use AI to classify maintenance requests.
- Automatically recommend problem severity.
- Detect potentially duplicate requests.
- Recommend suitable technicians.

- Allow users to track request progress.
- Allow technicians to maintain repair records.
- Allow users to verify completed repairs.
- Provide management analytics and historical maintenance data.
- Build a commercially viable B2B software product.

8. Scope

In Scope
The first version of FixFlow AI will include:

- User registration/login
- Role-based access
- Maintenance request creation
- Location selection
- Problem description
- Image upload
- AI problem classification
- AI severity recommendation
- Duplicate request detection
- Technician assignment
- Request status tracking
- Notifications
- Repair history
- User verification
- Manager dashboard

- Maintenance analytics

Out of Scope

To keep the project realistic and achievable, the first version will not include:

- Physical IoT sensors
- Automatic control of machines
- Robotics
- Fully autonomous technician assignment
- Advanced predictive-maintenance hardware
- Complex ERP integration
- Automatic purchasing of spare parts

These can be considered future extensions.

9. AI Component

AI Feature 1 — Problem Classification

The user enters:

"There is water leaking from the ceiling in Room 205."

AI predicts:

Category: Plumbing

Another example:

"The power socket is producing sparks."

AI predicts:

Category: Electrical
Severity: Critical

10. AI Feature 2 — Severity Recommendation

The AI analyzes the problem description and recommends:

Low
Example:

Broken chair.

Medium

Example:
AC not cooling.

High
Example:
Water leakage near electrical equipment.

Critical

Example:

Exposed electrical wiring producing sparks.

The manager can override the AI recommendation.

This is important because the system should assist the manager rather than blindly make decisions.

11. AI Feature 3 — Duplicate Detection

Suppose three employees submit:

"AC in Conference Room 2 isn't working."

"Conference Room 2 AC is not cooling."

"The air conditioner in Room 2 is broken."

The system can identify that these reports may refer to the same unresolved problem.

Instead of creating three completely separate tickets, the system can suggest:

"A similar unresolved maintenance request already exists."

The user can then follow the existing request.

12. AI Feature 4 — Technician Recommendation

Suppose the problem is classified as:

#Category: Electrical

The system checks:

- Technician skills
- Availability
- Current workload
- Location

and recommends:

Recommended Technician: Technician A

The manager can accept or change the recommendation.

13. Main Features

A. Maintenance Request

Users can:

- Report problem
- Select location
- Add description

- Upload photo
- Submit request

B. AI Analysis
System analyzes:

- Category
- Severity
- Possible duplicate
- Suggested technician

C. Manager Dashboard
Example:

FIXFLOW DASHBOARD

Total Requests	245
Open Requests	32
In Progress	18
Resolved	195

Critical	3
High	14
Medium	57
Low	171

D. Technician Dashboard
Technician sees:

MY ASSIGNED REQUESTS

#1021 Electrical Critical
#1028 HVAC Medium
#1034 Plumbing High

Each request can show:

- Location
- Description
- Image
- Priority
- Deadline
- Status

E. Status Tracking

A request follows:

Reported
↓
Under Review
↓
Assigned
↓
In Progress
↓
Resolved
↓
Verified

↓
Closed

If the customer rejects the resolution:

Resolved
↓
Reopened

14. Functional Requirements

Following the format your faculty provided:

ID	Requirement Type	Description	Priority
FR1	Functional	The system shall authenticate users according to their assigned roles.	Must
FR2	Functional	The system shall allow users to submit maintenance requests with location, description and optional images.	Must
FR3	Functional	The system shall use an AI component to suggest the category and severity of a maintenance request.	Must

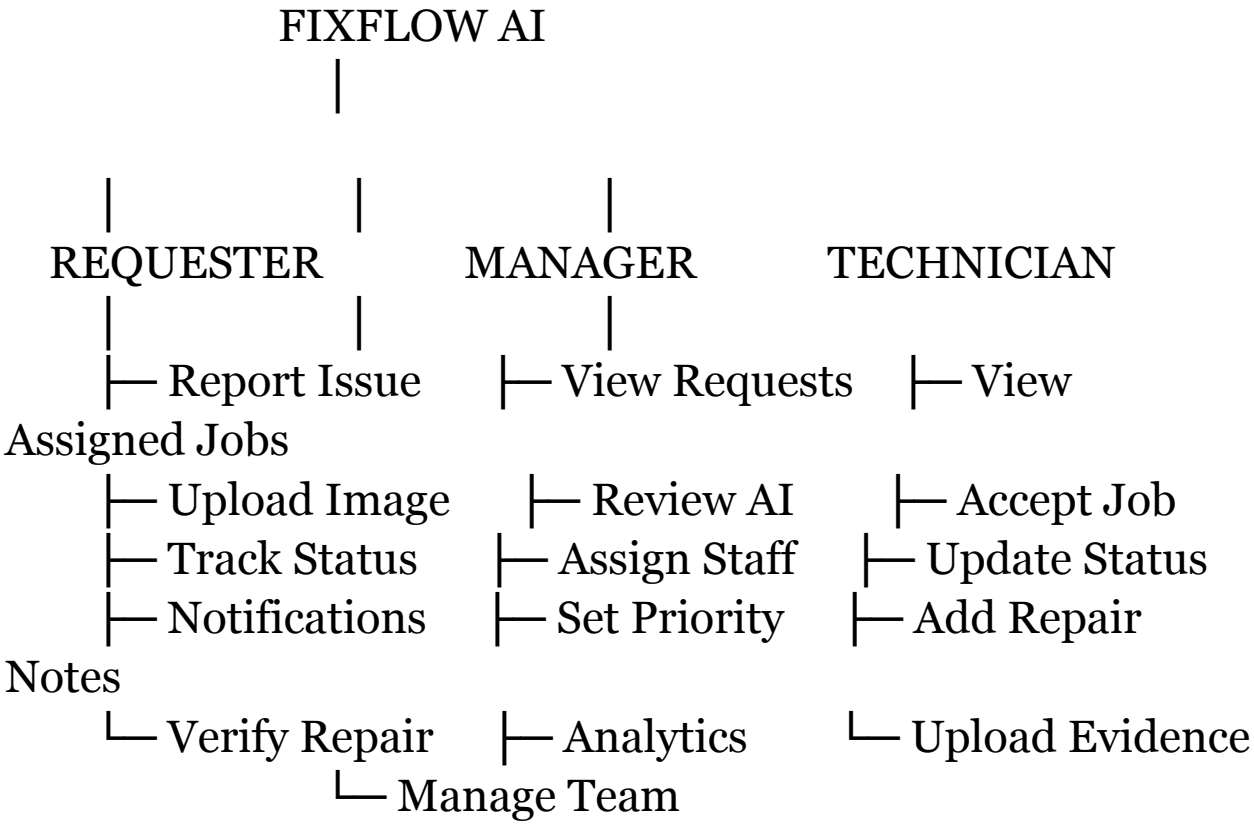
FR4	Functional	The system shall identify potentially duplicate unresolved maintenance requests.	Should
FR5	Functional	The system shall allow managers to assign maintenance requests to technicians.	Must
FR6	Functional	The system shall allow technicians to update request status and add repair information.	Must
FR7	Functional	The system shall allow requesters to verify completed maintenance requests and reopen them if the problem remains unresolved.	Should

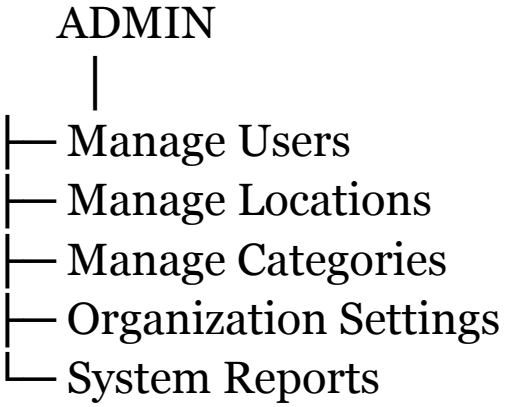
15. Non-Functional Requirements

ID	Requirement Type	Description	Priority
NFR1	Non-functional	The system shall normally respond to standard user requests within 2 seconds under the defined prototype load.	Must

NFR2	Non-functional	The system shall enforce role-based authorization for protected functions.	Must
NFR3	Non-functional	The system shall maintain an audit history of important request status changes.	Must
NFR4	Non-functional	The system shall provide a responsive interface for desktop and mobile browsers.	Should

16. Role-Wise Feature List





This can be converted into a proper diagram for your presentation.

17. User Stories

User Story 1 — Report Problem
As a requester, I want to submit a maintenance problem with its location and description so that the maintenance team can address it.
Acceptance Criteria
1.User must provide a valid location and description.
2.System generates a unique maintenance request ID.
3.The request appears in the user's request history.

User Story 2 — AI Analysis
As a manager, I want the system to suggest the category and severity of a maintenance problem so that urgent problems can be handled appropriately.

Acceptance Criteria

1.System generates a category and severity for supported descriptions.
2.Critical requests are clearly highlighted.
3.Manager can modify the AI recommendation.

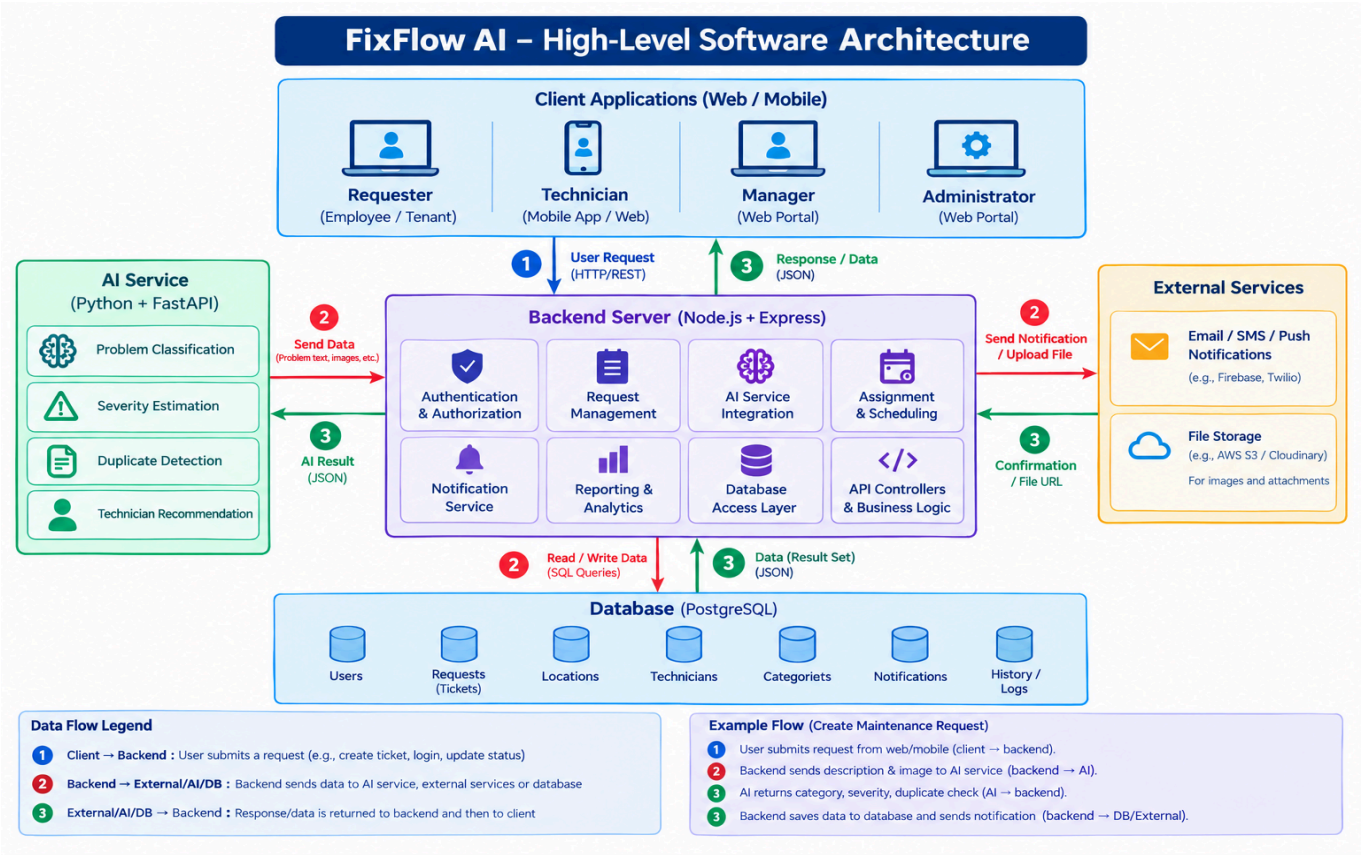
User Story 3 — Technician
As a technician, I want to see my assigned maintenance requests so that I can efficiently manage my repair tasks.

Acceptance Criteria

1.Technician sees assigned requests.
2.Technician can update request status.
3.Technician can add repair notes and upload evidence.

User Story 4 — Verification
As a requester, I want to verify whether the reported problem has actually been fixed so that unresolved problems can be reopened.
Acceptance Criteria
1.User can select "Fixed" or "Not Fixed."
2.Selecting "Not Fixed" reopens the request.
3.Manager receives a notification when a request is reopened.

18. High-Level Software Architecture



19. Technology Stack

Frontend

React.js

Used for:

- User interface
- Dashboards
- Request forms
- Manager interface
- Technician interface

Backend
Node.js + Express.js

Used for:

- REST APIs
- Authentication
- Business logic
- Request management
- Assignment
- Notifications

Database
PostgreSQL

Possible tables:

- Users
- Organizations
- Locations
- Technicians
- MaintenanceRequests
- Categories
- Assignments
- Attachments
- Notifications
- Feedback
- StatusHistory

AI
Python + FastAPI

Possible AI services:

Text
↓
Classification Model
↓
Category
Severity
Duplicate similarity

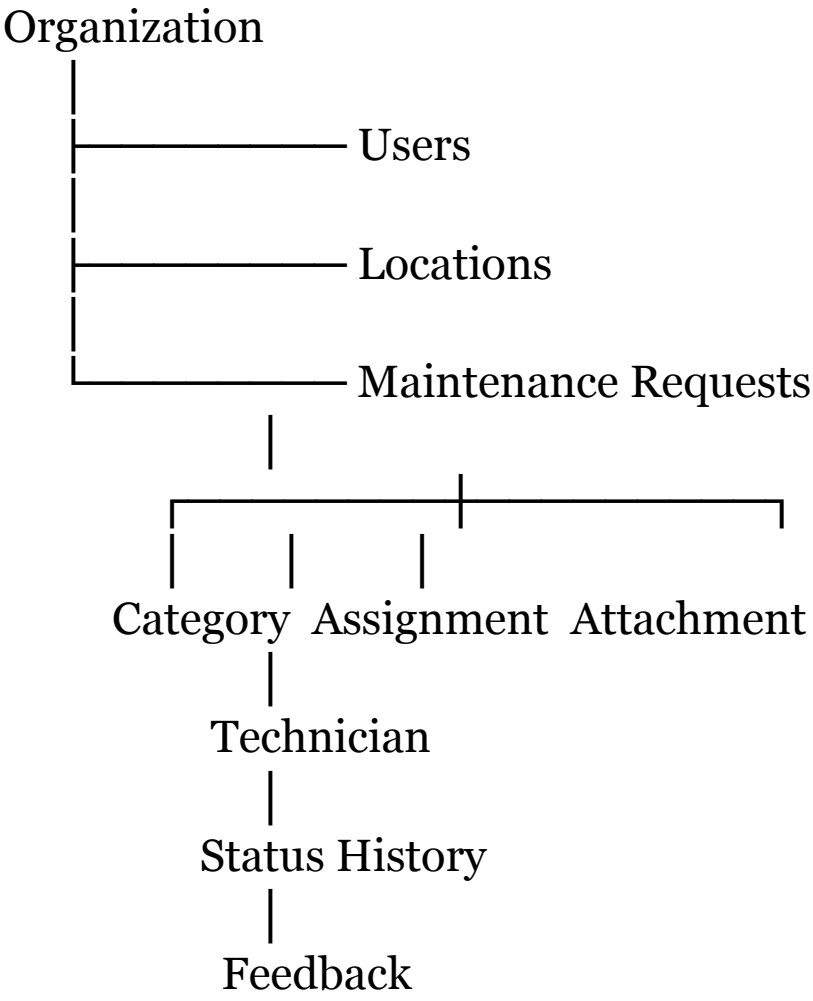
Version Control:
Git + GitHub

Deployment:
For the prototype:

- Frontend hosting
- Backend cloud hosting
- PostgreSQL cloud database
- AI service deployment

20. Database Concept

A simplified relationship could be:

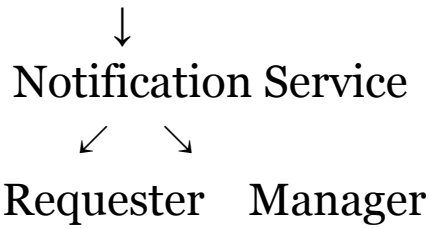


21. Possible Design Patterns

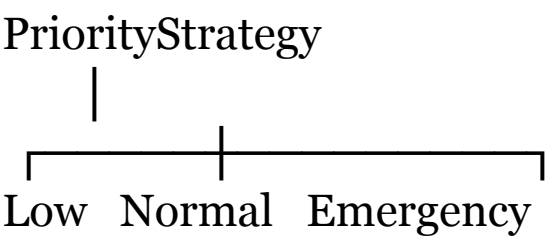
Our course includes design patterns, but our faculty's proposal doesn't require you to demonstrate every pattern now.
Still, FixFlow AI has natural places for the.

Observer Pattern
For notifications.

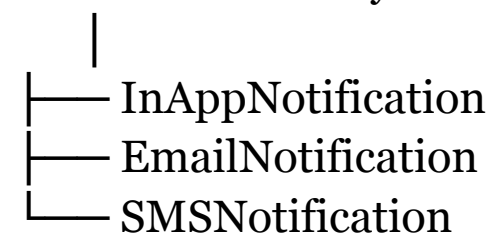
Ticket Status Changed



Strategy Pattern
For different priority-calculation strategies.



Factory Pattern
For different notification types:



These can later be documented in our design phase.

22. Risks

(Our proposal should include risks because our faculty explicitly asks for them.)

Risk	Impact	Mitigation
AI classification is inaccurate	Medium /High	Allow manager to modify AI results
Insufficient training data	High	Start with a limited set of maintenance categories and curated dataset
Scope becomes too large	High	Define MVP and postpone advanced features
Image uploads increase storage	Medium	Limit file size and use cloud storage
Unauthorized access	High	Role-based authorization and secure authentication
Poor user adoption	Medium	Keep reporting workflow simple
AI service unavailable	Medium	Allow manual category/severity selection

23. Business Model

Potential customers
A property-management company might have:

20 buildings + 500 employees/tenants + 15 technicians.

Instead of building their own software, they subscribe to FixFlow.

Example pricing model

Plan	Target
Starter	Small office/small building
Business	Medium organization
Professional	Large organization/multiple locations
Enterprise	Custom deployment

(We can use illustrative pricing in your presentation rather than claiming these are established market prices.)

Additional revenue could come from:

- Initial setup
- Customization
- Additional locations

- Premium AI features
- Data migration
- Technical support
- Custom reports
- Enterprise integration

24. Why Would a Customer Pay?

This should be one of your strongest presentation points.

A company isn't buying:

"a maintenance website."

It is buying:

Faster response
Problems reach the correct person faster.

Less downtime
Equipment problems can be addressed sooner.

Better accountability
Every request has an owner and status.

Better management
Managers can see maintenance performance.

Historical intelligence
The company can identify repeatedly failing equipment.

Reduced administrative work
Employees don't have to manage everything manually through messages and spreadsheets.

25. Similar Products

There are already commercial maintenance-management products, which is actually useful for your proposal because it demonstrates market validation.

For example, UpKeep is a commercial maintenance-management platform that provides work-order management and maintenance workflows. Its product materials also describe AI-assisted functionality.

Our proposal should not claim:

"No one has ever built this."

Instead:

"Existing products demonstrate demand for maintenance-management software. FixFlow AI proposes a focused and affordable solution for small and medium-sized organizations, with AI-assisted classification, severity detection, duplicate detection and technician recommendation."

That is a much more defensible novelty claim.

26. Proposed Novelty

Main novelty

FixFlow AI combines AI-assisted maintenance request classification, severity recommendation, duplicate detection and technician recommendation into one lightweight platform designed specifically for small and medium-sized organizations.

Additional differentiators

1. Explainable AI
Instead of only saying:

"Critical"

the system can show:

"Critical because the report contains indicators of electrical hazard."

2. Human-in-the-loop
Managers can override AI decisions.

3. Affordable
Designed for organizations that don't need a huge enterprise CMMS.

4. Flexible

The same platform can be configured for:

- Hotels
- Offices
- Schools
- Apartments
- Hospitals
- Factories

27. Proposed 8-Week Gantt Chart

Activity	W 1	W 2	W 3	W 4	W 5	W 6	W 7	W 8
Requirement Analysis	✓	✓						
UI/UX Design		✓	✓					
Database Design		✓	✓					
Backend Development			✓	✓	✓			
Frontend Development			✓	✓	✓			
AI Module				✓	✓	✓		
Integration						✓	✓	

Testing							✓	✓
Deployment							✓	
Documentation							✓	✓
Final Presentation								✓

28. Work Distribution

Member 1 — Project Lead + Backend
Responsibilities:
Requirements
Backend architecture
Authentication
APIs
Ticket management
Integration

Member 2 — Frontend Developer
Responsibilities:
UI/UX
Requester dashboard
Manager dashboard
Technician dashboard
Responsive interface

Member 3 — AI/Data Engineer
Responsibilities:
Dataset preparation
Text classification
Severity model
Duplicate detection
AI API

Remaining
Database + QA + DevOps
Responsibilities:
PostgreSQL
Database design
Testing
GitHub management
Deployment
Documentation
Security/performance testing

29. Why FixFlow AI Is a Good CSE327 Project

The important thing is that you don't need to implement every feature I listed.

For the MVP, you could focus on:

Authentication
↓
Report Request
↓
AI Classification
↓
Priority
↓
Manager Assignment
↓
Technician Update
↓
User Verification

Then we can add:(If we get extra time)

Duplicate Detection
Analytics
Notifications
Technician Recommendation